

Bioluminescence (BLI) & Chemiluminescence Imaging using Irradiators

Preclinical Imaging and Radiation Therapy with Precision X-Ray SmART+ and X-Rad Systems

This application highlights potent preclinical research solutions provided by the combination of bioluminescence and chemiluminescence imaging with Precision X-Ray's SmART+ and X-Rad systems, enhanced by the OptiMAX module. In addition to supporting accurate radiation delivery and enabling real-time monitoring of tumour dynamics and therapy responses, these sophisticated multimodal imaging system improves detection sensitivity and resolution. Researchers can evaluate the effectiveness of new medication therapies, look into changes in the tumour microenvironment, and improve radiotherapy procedures by combining the SmART+ platform with OptiMAX. Preclinical irradiators are more accurate thanks to bioluminescence imaging's capacity to visualise both anatomical and molecular features, which eventually advances cancer research and treatment development.

Keywords or phrases: Bioluminescence, chemiluminescence, preclinical imaging, radiation therapy, tumour microenvironment, image-guided radiotherapy, in vivo imaging, cancer, radiobiology, molecular imaging

Introduction

Bioluminescence and chemiluminescence are captivating phenomena involving the emission of light through biological processes and chemical reactions, respectively (Figure 1). Bioluminescence is the production and emission of light by living organisms, occurring through biochemical reactions involving luciferins (light-emitting molecules) and luciferases (enzymes that catalyze the reaction). Chemiluminescence refers to light emission resulting from a chemical reaction, occurring in both biological and non-biological contexts. Unlike bioluminescence, it is not exclusive to living organisms.

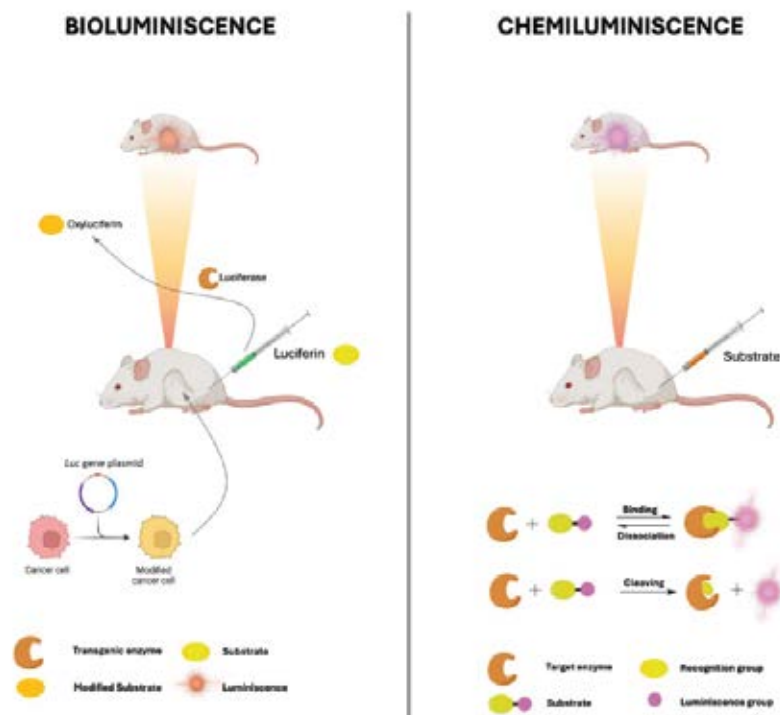


Figure 1: Overview of bioluminescence processes (left) and chemiluminescence processes (right).

The Precision X-Ray, Inc. SmART+ and X-Ray family are state-of-the-art preclinical irradiators, which can be enhanced with advanced imaging capabilities through the OptiMAX module, offering Optimized Detection, an Enhanced Imaging Workflow, and Advanced Software. The integration supports a wide range of applications, including Preclinical Imaging of Tumour Response to Radiation, Evaluating the Efficacy of Novel Drug Treatments, and Tumour Microenvironment and Therapeutic Responses, as described in examples below-

1. Radiotherapy-Induced Astrocyte Senescence Promotes an Immunosuppressive Microenvironment in Glioblastoma to Facilitate Tumour Regrowth

In this study¹, mice were cranially irradiated with a single-fraction dose of 10 Gy using the X-RAD225 OptiMAX (manufactured by Precision X-Ray) at 225 kV, 13.3 mA, 2.05 Gy/min with a 2-mm aluminium filter.

The device was equipped with a custom collimator to deliver a 20 mm-diameter iso-dose field while shielding the rest of the body. This irradiation setup was critical for inducing radiation-induced senescence in astrocytes, which were later found to promote an immunosuppressive tumour microenvironment facilitating glioblastoma regrowth. Thus, the SMART+ platform was a core component in enabling the study of radiation effects *in vivo*.

2. NRP1 inhibition modulates radiosensitivity of medulloblastoma by targeting cancer stem cells

Precision X-Ray was utilized to enhance the accuracy of radiation therapy in medulloblastoma (MB) models². By combining precise radiation delivery with the inhibition of Neuropilin-1 (NRP1) using the peptidomimetic agent MR438, the researchers were able to selectively target tumour cells while minimizing damage to surrounding healthy tissue. This precise modulation of radiation

Application Showcase

allowed for improved tumour control and a significant reduction in cancer stem cells, providing a promising strategy for reducing radiation doses while enhancing therapeutic efficacy in MB treatment.

3. Imaging Performance of a Multimodal Module to Enhance Preclinical Irradiator Capabilities

Precision X-Ray's OptiMAX imaging system was integrated into a preclinical irradiator to enhance imaging capabilities for both X-ray and bioluminescence applications³. The system, which uses a cooled CCD camera and a phosphor screen, allows for high-resolution imaging by capturing radiographic and optical images with a single sensor. The X-ray images demonstrated excellent spatial resolution (2.5-line pairs per millimetre) and high contrast sensitivity, while bioluminescence imaging provided detailed insights into the activity of cancer cells in vivo. This multimodal imaging approach improves the preclinical irradiator's ability to conduct image-guided radiotherapy, offering better anatomical and molecular visualization, essential for optimizing radiotherapy protocols in animal models.

Conclusion

The integration of Bio/Chemiluminescence Imaging (BLI) with the Precision Smart+ and XRAD systems, enhanced by OptiMAX, marks a significant improvement in the imaging capabilities of these systems. This integration facilitates targeted radiation delivery and allows for effective monitoring of processes and their efficiency. Additionally, these imaging capabilities can be employed independently of radiation for research purposes. This combination enhances preclinical research sensitivity, resolution, and workflow efficiency, facilitating a broad spectrum of applications in preclinical research.

References

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PRECISION
X-RAY IRRADIATION



X-RAY SmART+